Digital CAD Training and Services

NX 12 45 Days Syllabus

Day-wise learning plan with practical tasks, project work and final assessment

Duration	Software	Training Focus
45 Days	NX 12	Sketching, Modeling, Synchronous, Surface, Assembly, Drafting, Project

This document is prepared for students to clearly understand the complete NX course journey, daily topics, practice expectations, project work and final outcome before joining the training.

Includes CADPOINT Authorized Training Centre logo as requested.

Course Objective

This syllabus is designed to build a strong practical foundation in NX 12 through a clear 45-day progression. Students begin with sketching fundamentals, move into feature-based modeling and synchronous editing, continue with surface creation and analysis, learn assembly and drafting workflows, and complete the course with project work and final drawing assessment.

What Students Will Learn

- Create fully defined NX sketches using geometric and dimensional constraints.
- Build 3D parts using extrude, revolve, hole, blend, chamfer, draft, shell, rib, pattern and mirror tools.
- Use synchronous modeling tools such as Move Face, Resize Face, Delete Face and Replace Face.
- Create and edit surface features including trim sheet, sew, offset and surface analysis.
- Prepare assemblies using assembly navigator, constraints, touch/align and exploded views.
- Create production drawings with standard views, sections, dimensions, tolerances, GD&T;, parts list and balloons.

Module Breakdown

Module	Days	Focus Area
Sketch	**5**	2D profile creation, sketch constraints, dimensions and sketch validation
Modeling	**10**	Core 3D feature creation, editing and feature management
Synchronous	**5**	Direct geometry editing and topology modification
Surface	**8**	Surface generation, trimming, sewing, offset and analysis
Assembly	**5**	Component insertion, constraints and exploded view creation
Drafting	**9**	Drawing setup, views, dimensions, tolerances, GD&T, parts list and templates
Project	**3**	Mini project execution, final drawing preparation and assessment

Complete Day-wise Syllabus

Each day includes one focused topic and one daily practice task. Students should complete the practice task on the same day to maintain learning continuity and improve software command.

Day	Module	Topic	Daily Practice Task
1	Sketch	NX Interface + Navigation	Create basic plate sketch
2	Sketch	Sketch Tools - Line, Circle, Arc	Create bracket profile
3	Sketch	Geometric Constraints	Apply constraints
4	Sketch	Dimensional Constraints	Modify sketch
5	Sketch	Sketch Analysis	Fully constrain complex sketch
6	Modeling	Extrude	Create base solid
7	Modeling	Revolve	Create cylindrical part
8	Modeling	Hole Command	Create hole features
9	Modeling	Edge Blend	Apply blends
10	Modeling	Chamfer	Apply chamfers
11	Modeling	Draft	Apply draft
12	Modeling	Shell	Hollow part
13	Modeling	Rib	Create reinforcement
14	Modeling	Pattern Feature	Create patterns
15	Modeling	Mirror Feature	Mirror features
16	Synchronous	Move Face	Modify geometry
17	Synchronous	Resize Face	Adjust model
18	Synchronous	Delete Face	Edit topology
19	Synchronous	Replace Face	Modify surfaces
20	Synchronous	Direct Edit	Edit model directly
21	Surface	Extrude Surface	Create surface
22	Surface	Revolve Surface	Create revolve surface
23	Surface	Sweep	Create sweep
24	Surface	Through Curve Mesh	Create mesh surface
25	Surface	Trim Sheet	Trim surfaces
26	Surface	Sew	Join surfaces
27	Surface	Offset Surface	Create offset
28	Surface	Surface Analysis	Check quality
29	Assembly	Assembly Navigator	Insert parts
30	Assembly	Assembly Constraints	Apply constraints
31	Assembly	Touch / Align	Align components
32	Assembly	Angle Constraint	Apply angle
33	Assembly	Exploded View	Create explode
34	Drafting	Drafting Setup	Create drawing

Day	Module	Topic	Daily Practice Task
35	Drafting	Standard Views	Generate views
36	Drafting	Section View	Create section
37	Drafting	Dimensioning	Add dimensions
38	Drafting	Tolerances	Apply tolerances
39	Drafting	GD&T	Apply GD&T
40	Drafting	Sheet Setup	Prepare final sheet
41	Drafting	Parts List and Balloons	Create parts list and add balloons to assembly drawing
42	Drafting	Title Block and Drawing Template	Customize title block and save drawing template
43	Project	Mini Project - NX Modeling	Create one complete mechanical part using sketching, modeling and synchronous tools
44	Project	Mini Project - Assembly	Assemble project components, apply constraints and create exploded view
45	Project	Final Drawing and Assessment	Prepare final drawing sheets and complete final evaluation submission

Student Practice Guidelines

- Complete the daily practice task before moving to the next day.
- Keep every file properly named with day number, topic and your name.
- Maintain separate folders for Sketch, Modeling, Synchronous, Surface, Assembly, Drafting and Project files.
- Revise previous commands weekly to improve speed, confidence and command retention.
- Submit doubts with screenshots, file name and the exact step where you are facing the issue.
- Complete the mini project and final assessment before claiming course completion.

Final Outcome After 45 Days

Area	Expected Outcome
Sketching	Students can create and modify constrained NX sketch profiles confidently.
Modeling	Students can build practical 3D parts with common production features.
Synchronous	Students can directly edit model faces and modify geometry efficiently.
Surface	Students can create, trim, sew and analyze surface geometry.
Assembly	Students can insert components, apply constraints and create exploded views.
Drafting	Students can prepare drawings with views, dimensions, tolerances, GD&T, parts list, balloons and title blocks.
Project	Students can complete a mini project and submit final drawing sheets for assessment.

For course details and batch updates, contact Digital CAD Training and Services.